Home Stretch

European Housing Affordability Dashboard

Mapping Europe's housing cost squeeze from country trends to city–rural differences.

WEEK 1 PROJECT • DATA ANALYSIS WITH VIBE CODING • OCTOBER 2026

Github Link: https://github.com/anshulgoyl/AI-Course-week1/tree/main

Doc: Home_Stretch_Week_1_Project_Summary

Demo Link: https://www.loom.com/share/723cfdbd99bc4821b863bdb11509a198

1. What am I building

Project Inspiration. I built this dashboard to make a complex affordability statistic easier to explore: where housing pressure is highest, how it has changed since 2014, compare multiple countries together and whether the experience differs between cities, towns and rural areas.

Housing cost overburden rate: the percentage of people living in households where total housing costs exceed 40% of disposable household income.

Home Stretch is an interactive European housing affordability dashboard. It translates Eurostat's housing cost overburden rate into visuals, users can begin with the European picture, select a country, compare its history, and examine how pressure differs across its urban and rural geographies. Together, these views help distinguish a temporary change from a long running structural pattern.

2. Data source & overview

The core measure is the housing cost overburden rate: the share of people living in households where housing costs exceed 40% of disposable income. The dashboard combines a national time series with a separate Eurostat urbanisation extract so that a country total is not mistaken for the experience of every place within that country.

Source	What it contains	How I used it
Eurostat estat_tespm140.tsv	National housing cost overburden rates, 2014–2025.	Powers the European map, the rankings, the country trends and the archetypes.
Eurostat tessi165 / ilc_lvho07d	2025 overburden rates by degree of urbanisation: cities, towns/suburbs and rural areas.	Powers the selected country's city–rural comparison in the deep dive.

Coverage used in the dashboard: annual national trends from 2014 to 2025, with the 2025 urbanisation breakdown shown for the actively selected country. Values are displayed as percentages and include Eurostat estimate, provisional and break in series flags where supplied.

3. Tools used

- Codex: Used as an AI coding collaborator to interpret the files, build the interface, debug interactions and refine the explanatory text.
- HTML, CSS and vanilla JavaScript: Used to build a lightweight browser dashboard with fixed top navigation and interactive charts.
- Eurostat data extracts and GeoJSON country boundaries: Used for the measures, time series, urbanisation comparison and clickable map.
- GitHub: Used to create a repository for the final code and data assets.

4. Starting point

I began with a simple decision question: where is Europe's housing cost squeeze most acute, and is a national average telling the whole story? The initial task was to make the country level overburden series usable; the second was to connect it to a spatial view and the urbanisation breakdown without losing clarity.

This shaped the design. The map is the starting point, not decoration: a click changes the country trend on the same screen. The deep dive then separates national movement from the 2025 geography within the selected country, helping users see whether pressure is concentrated in cities, spread across settlement types, or unexpectedly stronger outside them.

5. Iterations involved

The dashboard improved through short, testable iterations rather than one final design pass:

- Data reading: Validated the national TSV and re read the Excel extract to recover the city, town/suburb and rural breakdown needed for the deep dive.
- Map repair: Replaced a schematic country layout with actual European country boundaries and connected each country shape to the trend selection state.
- Comparison clarity: Expanded the country ranking into a full horizontal bar chart and added small direction indicators so the latest level and recent movement can be scanned together.
- Dynamic interpretation: Rewrote the explanatory copy around the selected country's actual trend and urbanisation profile instead of generic next step advice.
- Pattern view: Developed four interpretable market archetypes and added a shared trend chart for the countries in the selected archetype.

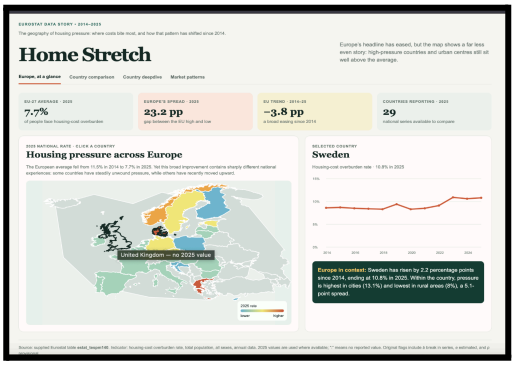
Prompts Sequence

1. **Define the dashboard vision:** Build an interactive dashboard called "Home Stretch" to visualise housing cost overburden across Europe. Start with a 2025 European map, show Europe wide KPIs, and allow users to move from the European picture into country trends and city versus rural differences.
2. **Design the four tab experience:** Create four fixed top navigation tabs: Europe at a glance, Country comparison, Country deepdive, and Market patterns. Keep the tabs visible while scrolling, use a clean visual hierarchy, improve chart readability, and add data source footnotes across every tab.
3. **Read and integrate the data:** Read the supplied Eurostat TSV and Excel files correctly. Use the national 2014–2025 series for country trends and rankings, and use the Excel based urbanisation data for city, town/suburb, and rural comparisons. Ensure the urbanisation view updates for the selected country.
4. **Build and repair the European map interaction:** Use a real Europe country boundary map rather than a schematic layout. Colour countries using 2025 housing cost overburden data. When a user clicks a country, update the trend chart on the right and carry that selection into the country deep dive view.
5. **Develop country level analysis and comparisons:** Create a full horizontal ranking of countries and add up, down, or flat icons to show the long run direction of each country's trend. Include a multi country trend explorer so users can compare two to five countries and distinguish persistent pressure, improvement, and recent deterioration.

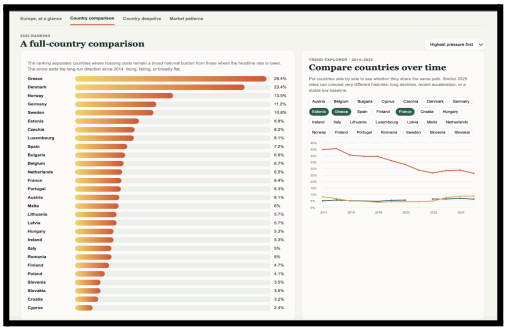
6. Final outcome

The finished result is a four tab decision dashboard titled Home Stretch. Its fixed top navigation keeps the views accessible, while the content moves deliberately from Europe wide context to a country comparison, a country specific place profile and a set of comparable market patterns.

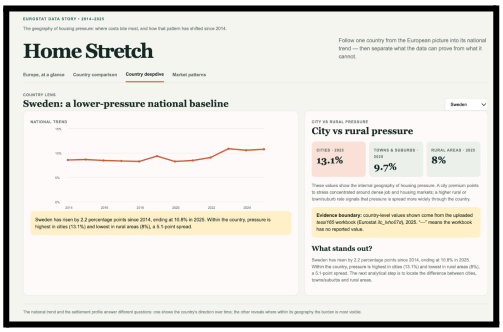
What a reader can see. The experience now combines the level of housing pressure in 2025, its direction since 2014, and its distribution across cities, towns/suburbs and rural areas. That makes the dashboard useful for both a citizen trying to understand a place and a policy reader looking for differences that a European average conceals. There are 4 key visual outputs to understand the European Housing Affordability Dashboard.



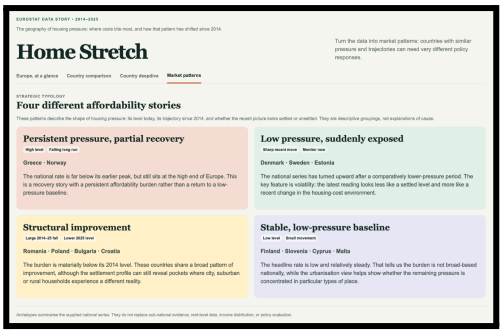
Europe at a glance: This landing view gives a Europe wide picture of housing cost pressure in 2025. It combines headline KPIs with a clickable country boundary map and a linked trend chart. The European average has eased from 11.5% in 2014 to 7.7% in 2025, but the map makes clear that this average hides major differences between countries. Selecting Sweden, for example, reveals a recent upward trend and a higher city level burden than its national average suggests.



Country comparison: This view ranks countries by their 2025 housing cost overburden rate, making the cross country spread immediately visible. Greece and Denmark appear as the highest pressure markets, while several Central and Eastern European countries sit at much lower levels. The accompanying trend explorer lets users compare multiple countries from 2014 to 2025. It shows why similar looking current rates can reflect very different stories, such as long term recovery, stable low pressure, or a recent increase.



Country deepdive: The country deep dive connects Sweden's national trend with its internal geography. Sweden's overburden rate increased from 8.6% in 2014 to 10.8% in 2025, indicating a gradual rise in national pressure. The settlement breakdown shows that pressure is highest in cities at 13.1%, compared with 9.7% in towns and suburbs and 8.0% in rural areas. This suggests that the headline national figure masks a clear urban affordability premium.



Market patterns: The market pattern view groups countries into four descriptive affordability archetypes using their current pressure level and direction of change since 2014. Greece and Norway show persistent pressure with partial recovery, while Denmark, Sweden and Estonia are lower pressure markets with recent upward movement. Romania, Poland, Bulgaria and Croatia show structural improvement, while Finland, Slovenia, Cyprus and Malta form a stable, low pressure baseline. These groups help compare trajectories, not assign causes.

7. Key learnings

- **Validate the data before designing the dashboard.** The national time series and urbanisation data came in different file structures and answered different questions. Checking definitions, years, missing values, and geography levels was essential before combining them.
- **Design the analysis as a journey from broad to specific.** The dashboard moves from Europe-wide patterns to country rankings, individual trends, and city, town, suburb, and rural differences. This helps users understand context before interpreting detail.
- **Make every interaction purposeful.** The map became valuable when selecting a country and updating the linked trend and deep-dive views. Interactive features should help users answer a follow-up question, not simply make the dashboard look more advanced.
- **Use AI through precise, testable prompts.** Focused feedback such as repairing the map, linking country selection, adding trend signals, or rewriting generic insight text produced stronger results than broad requests for improvement.
- **Iterate in small steps and verify each result.** Data values, map boundaries, calculations, written insights, and click behaviour all needed checking against the source files. AI accelerated the build, but validation remained the user's responsibility.
- **Pair visuals with interpretation.** Charts communicate the pattern, while concise supporting text explains why the pattern matters. The best insights highlighted differences between current pressure, long-term change, and variation within a country.

This project showed me how quickly a raw public dataset can become a decision ready story when data validation, interaction design and iterative prompting are treated as one workflow.